London Transit Resilience System (LTRS)

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Project URLs

Resource	Link
Final Assignment URL (live website)	https://yuxiang2526.github.io/ltis-london-transit-immune-system/
Project code files (web frontend)	https://github.com/Yuxiang2526/ltis-london-transit-immune-system
Interactive Map · code repository	https://github.com/Taoo2025/CASA0029
Data and Data-Processing Code Files	https://github.com/Taoo2025/CASA0029/tree/main/data_calculating

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Project Output Table

#	Output	Location	Description
1	**Story** (scrollytelling narrative)	`/` route on the live site	Editorial three-act walk-through of the LTRS findings using routes 99, R2 and 685, with per-act camera fly-to and bolded key statistics.
2	**Explorer** (LSOA narrative dashboard)	`/explore` route	Routes ranking + Impact Matrix + LSOA baseline choropleth with hover-to-inspect Local Profile + Lorenz curve / Decile bars / Distribution panel + Key Findings + Planning Implications.

#	Output	Location	Description
3	**Network Map** (forensic 100 m grid tool)	`/network` route	Full Mapbox GL JS map with Scenario Builder; user can cancel any combination of 543 routes and inspect the recomputed AI loss at native 100 m grid resolution; baseline / disrupted split-screen with draggable divider.
4	**Methodology**	`/methodology` route + `docs/methodology.md`	Five-section methodology summary (Introduction, Data, Methodology with formulas, Discussion & limitations, References).
5	**About**	`/about` route	Site structure, contributor portraits, public data sources, honest caveats about what the project does NOT do.
6	**Reproducible code**	`web/` (frontend) + companion repo `Taoo2025/CASA0029/data_calculating` (pipeline)	All scripts, data and configuration are open source under MIT licence (web) / public access (data pipeline).

Methodology Summary

1. Introduction

In complex urban systems, the stability of public transport networks affects not only daily commuting efficiency, but also residents' access to jobs, education, healthcare and public services. Previous studies have shown that transport resilience is an important foundation for the functioning of urban society (Chopra et al., 2016). At the same time, transport equity research has gradually moved beyond simple measures of transport supply or passenger capacity. It now pays more attention to whether different areas and social groups can access travel opportunities fairly (Li et al., 2025). Recent studies have also compared accessibility differences across education groups, transport modes and job opportunities, showing that urban transport systems can produce both spatial and social inequalities (Liu and Yu, 2025).

However, most existing studies focus on accessibility under normal operating conditions. Less attention has been paid to whether different areas can maintain their original accessibility when the transport network is disrupted. For cities with complex public transport systems, high accessibility in normal conditions does not necessarily mean strong transport resilience. If an area depends heavily on a small number of key routes or stations, its accessibility may drop quickly when these links are disrupted. Therefore, transport equity should not only ask *who has better accessibility in normal conditions*. It should also ask *who is more likely to lose accessibility after a disruption*. Based on this research gap, this project extends the focus from static accessibility distribution to accessibility retention under route disruption scenarios. In this project, this retention capacity is defined as **transport resilience**.

2. Data

This project uses multi-source open datasets to assess London's public transport resilience at two spatial scales: 100 m grids and LSOAs.

1. **2023 PTAL dataset** — informs the baseline accessibility layer and measures the quality of public transport connectivity in London.
2. **Public transport access points** — derived from Great Britain's national NaPTAN dataset; London transport station and route data are used to represent local services. Night buses and temporary bus routes are excluded to mitigate the impact of irregular bus services on travel resilience. The GLA's Statistical GIS Boundary Files are used as a spatial mask to retain only transit data falling within the London administrative area.
3. **London network topology data** — supports the modelling of route disruption scenarios and accessibility change.

3. Methodology

The core contribution of this model lies in the development of an interactive platform supporting "scenario analysis." Diverging from the static PTAL data provided by TfL's official WebCAT, this model implements real-time re-computation within the web browser. The specific operational procedures are as follows:

1. **The baseline map is built on the 2023 PTAL values.** It visualises the current inequality in public transport connectivity shown by the official PTAL data.

2. **Following TfL's PTAL calculation logic**, the map estimates accessibility for each 100 m × 100 m grid using the real OSM walking network and open public transport stop data. For each grid centroid, reachable stops are identified within PTAL walking catchments: 640 m for bus stops and 960 m for rail-based stops. These catchments are measured along the OSM walking network with a walking speed of 4.8 km/h. The Accessibility Index is calculated by summing the contributions of all reachable routes:

   $$\mathrm{AI}_i = \sum_{s \in S_i} \sum_{r \in R_s} C_{isr}$$

   where S_i is the set of reachable stops from grid i, R_s is the set of routes serving stop s, and C_{isr} is the accessibility contribution of route r at stop s. The route contribution uses a distance-decay function:

   $$C_{isr} = w_{isr} \cdot \max\left(0,\ 1 - \frac{d_{is}}{D_m}\right) \cdot 10$$

   where w_{isr} is the route weight, d_{is} is the OSM-network walking distance from grid i to stop s, and D_m is the mode-specific catchment distance. A closer stop gives a higher contribution, while a stop outside the catchment gives zero contribution.

   Next, the average reachability is calculated using LSOA-based polygonal spatial aggregation for macro-level analysis.

3. **When the user cancels selected routes**, the map recalculates grid-level accessibility by removing the AI contributions of those routes. The accessibility loss is measured as:

   $$\mathrm{AI}_{\mathrm{loss},\,i} = \mathrm{AI}_{\mathrm{baseline},\,i} - \mathrm{AI}_{\mathrm{disrupted},\,i}$$

   and the **Local Transit Resilience Score** is calculated as:

$$\mathrm{LTRS}_i = \frac{\mathrm{AI}_{\mathrm{disrupted},\, i}}{\mathrm{AI}_{\mathrm{baseline},\, i}}$$

A larger accessibility drop indicates weaker resilience. Therefore, the lower the LTRS value, the more vulnerable the grid is under the selected route disruption scenario.

4. Discussion

Despite its effectiveness, this study has several limitations due to computational constraints. First, our model does not fully replicate the official TfL PTAL methodology. Specifically, we did not include frequency attenuation or multi-modal transfers, such as moving from a bus to the Underground. Furthermore, our analysis assumes a static environment and does not account for service changes during weekends or peak hours. Future research could improve these calculation methods to provide a more realistic simulation. It is also important to integrate more socio-demographic factors, such as the Index of Multiple Deprivation (IMD). By performing an IMD overlay analysis, researchers can better identify which vulnerable groups are most affected by transport disruptions. This would help planners understand how transport resilience contributes to wider social inequality in London.

References

- Chopra, S. S. et al. (2016) "A network-based framework for assessing infrastructure resilience: a case study of the London metro system", *Journal of The Royal Society Interface*, 13(118), p. 20160113.
- Li, A. et al. (2025) "Ease and Equity of Point of Interest Accessibility via Public Transit in the U.S". *arXiv*.
- Liu, Z. and Yu, Z. (2025) "Transport equity assessment based on accessibility disparities in terms of multi-job opportunities across Beijing", *Scientific Reports*, 15(1), p. 30878.

Appendix: Group Contribution and AI Declaration

1. Individual Contributions

As a two-person team, we collaborated closely throughout the project lifecycle. The workload was distributed equally, and the partnership was highly effective.

Contributor	Role
Siyan Tao	Lead on **data engineering, geospatial preprocessing, and the development of the interactive Mapbox engine**. OSM walking-network extraction, per-100 m-grid Accessibility Index computation, 543-route per-cell loss matrix pre-computation, and the Mapbox GL JS Network Map UI (split-screen compare, route-cancel logic, postcode/borough navigation).

Contributor	Role
Yuxiang Fan	Lead on **front-end web architecture, UI/UX design, and the drafting of the methodology and urban analysis summary**. React + TypeScript + Vite skeleton, design system, scrollytelling narrative, LSOA aggregation views, methodology and About pages, GitHub Pages deployment.

2. AI Tool Declaration

In line with the academic integrity policy for this module, we declare the use of the following Artificial Intelligence tools.

- **Code development.** The team designed the main logic and system structure. Some modular code components were generated and refined with Vibecoding models. Claude and Codex were also used to help with debugging and code optimisation.
- **Conceptual design.** The research questions, the definition of LTRS (London Transit Resilience Score), and the scenario analysis method were developed by the team. AI tools were only used as brainstorming support during discussion.
- **Editing.** Gemini was used to check grammar and improve the clarity of the final methodology summary.

The final content, interpretation, and project decisions remain the responsibility of the team.